

SALONE HEALTH WATCH

A Digital Public Good for Community Health Awareness in Sierra Leone

License: MIT Open Source

1. Introduction

Sierra Leone faces significant public health challenges, including high rates of malaria, cholera, and maternal mortality. Access to timely, accurate health information — especially in rural and underserved communities — can be the difference between early treatment and preventable death.

Salone Health Watch is a free, open-source Digital Public Good (DPG) designed to close this gap. Built entirely with standard web technologies (HTML5, CSS3, and JavaScript), it requires no internet connection after download, no server, and no installation — making it suitable for low-cost mobile devices and offline environments across Sierra Leone.

This project was developed for DLA207 — I.T Law and IPR Legal Issues, applying legal and ethical principles in software development including open licensing, data privacy, and accessibility.

1.1 Problem Statement

Many Sierra Leoneans, particularly in rural areas, lack access to reliable health guidance. Common illnesses like malaria and cholera are often not recognised early, leading to delayed treatment and worse health outcomes. Existing digital health tools often require continuous internet access, are not available in local languages, and collect personal data without clear user consent.

1.2 Our Solution

Salone Health Watch addresses these challenges through:

- A plain-language interactive symptom guide that works offline and collects no data
- Health topic cards covering the six most common health concerns in Sierra Leone
- A scrolling alert ticker for seasonal health messages (e.g. malaria season warnings)
- A rights and legal section explaining the MIT licence and data privacy to users
- SDG alignment clearly labelled throughout the tool

2. Project Objectives

This project was guided by the five objectives set out in the DLA207 project brief:

1. Design a functional digital solution addressing a real-world problem — the tool addresses healthcare access and early illness recognition in Sierra Leone.
2. Apply legal and ethical frameworks in software development — the MIT license is applied, user rights are explained in-app, and ethical data handling is built into the architecture.
3. Use an approved open-source license — the MIT License was selected and is included as a standalone LICENSE file.
4. Ensure data accessibility, interoperability, and privacy compliance — all health content is structured as JSON-like JavaScript objects; zero personal data is collected.
5. Develop skills in collaboration, documentation, and presentation — the team used GitHub for version control, produced this documentation, and prepared a presentation script.

3. Legal and Ethical Framework

3.1 Open Source License: MIT License

We selected the MIT License for this project for the following reasons:

- **Permissiveness:** It allows anyone — individuals, health organisations, schools, and developers — to use, copy, modify, and distribute the tool, including for commercial purposes.
- **Simplicity:** The license text is short and easy to understand, which aligns with our goal of making the tool accessible to non-technical stakeholders.
- **Compatibility:** MIT is compatible with most other open-source licenses, enabling integration with other systems without legal conflicts.
- **Wide adoption:** It is one of the most recognised and trusted open-source licenses globally, accepted by the Digital Public Goods Alliance (DPGA).

The LICENSE file is included in the project repository and the copyright notice is displayed in the application footer. The license text reads: "Permission is hereby granted, free of charge, to any person obtaining a copy of this software..." (full text in LICENSE file).

3.2 Data Privacy and Legal Compliance

Salone Health Watch was designed with privacy as a foundational principle, not an afterthought:

- **No personal data is collected.** The symptom guide runs entirely in the user's browser using client-side JavaScript. No information about selected symptoms is transmitted to any server.
- **No cookies or tracking scripts are used.** There are no analytics, advertisement scripts, or session identifiers.
- **No user accounts or registration.** The tool is fully accessible without any sign-up.
- **Offline capability.** Once downloaded, the tool works without an internet connection, removing any risk of data interception during transmission.

These principles align with international data protection standards including the General Data Protection Regulation (GDPR) principles of data minimisation and purpose limitation, even though Sierra Leone is not yet a GDPR jurisdiction.

3.3 Digital Public Goods Alignment

A Digital Public Good (DPG), as defined by the UN Secretary-General's Roadmap for Digital Cooperation and the Digital Public Goods Alliance, is an open-source

software, open data, or open content that adheres to privacy, accessibility, and other applicable best practices. Salone Health Watch meets the DPG criteria:

- Open licence: MIT License
- Does not collect or store personally identifiable information
- Documented: clear README and project documentation
- Platform independence: runs on any modern browser
- Non-discriminatory: freely available to all without restrictions based on who the user is

3.4 Ethical Considerations

- Medical disclaimer: The tool clearly states it provides general guidance only and is not a diagnostic tool. Every result panel includes a disclaimer directing users to seek professional care.
- Accuracy: Health content was reviewed against guidance from the World Health Organisation (WHO) and the Sierra Leone Ministry of Health and Sanitation.
- Non-alarmist language: Content was deliberately written to inform without frightening users, reducing the risk of panic or self-medication.
- Accessibility: The tool includes keyboard navigation, ARIA labels, screen reader-friendly markup, and respects the browser's reduced-motion setting — recognising that not all users have the same abilities or devices.

4. Technical Implementation

4.1 Technology Stack

The project uses three standard, dependency-free web technologies:

Technology	Version/Standard	Purpose
HTML5	HTML Living Standard	Document structure, semantic markup, accessibility attributes
CSS3	CSS3 Custom Properties	Responsive layout, animations, design tokens, dark/light sections
JavaScript	ES6 (Vanilla)	Symptom guide logic, DOM manipulation, event handling
Google Fonts	CDN (online only)	Typography: Archivo Black, Work Sans, JetBrains Mono

4.2 System Architecture

The tool is a single-page static web application. Its architecture is intentionally simple to maximise reliability and accessibility:

- Single file deployment (index.html) — originally combined, now separated into three files for maintainability: index.html (structure), style.css (presentation), script.js (behaviour).
- Client-side only — no backend server, no database, no API calls. All logic executes in the user's browser.
- Zero dependencies — no JavaScript frameworks (no React, Vue, or Angular). This keeps the file size small and eliminates dependency vulnerabilities.
- Offline-first — works without any internet connection after the initial download. Google Fonts gracefully degrades to system fonts if offline.

4.3 Key Features

4.3.1 Health Alert Ticker

A continuously scrolling banner across the top of the page displays contextual health alerts (e.g. "RAINY SEASON: MALARIA CASES RISE — SLEEP UNDER TREATED NETS"). This feature respects the prefers-reduced-motion CSS media query, disabling the animation for users who have indicated this preference in their device settings.

4.3.2 Health Topic Cards

Six topic cards cover the most prevalent health concerns in Sierra Leone: Malaria, Cholera & Water Safety, Maternal & Child Health, Immunisation, Hygiene & Sanitation, and Outbreak Alerts. Each card is tagged with its relevant SDG number. Content is structured as plain HTML, making it easily extractable and reusable.

4.3.3 Interactive Symptom Guide

The signature feature of the tool. Users select from eight common symptoms. The JavaScript logic applies a priority-based decision tree to determine the most appropriate guidance category:

Priority	Category	Triggered When
1 (Highest)	URGENT — Breathing	Difficulty breathing selected
2	URGENT — Dehydration	Dehydration signs selected
3	MONITOR — Pregnancy	Pregnancy + any other symptom
4	MONITOR — Rash	Rash with fever selected
5	MONITOR — Fever	Fever or severe headache selected
6	LOW — Mild	Any other symptom(s) selected
7 (Default)	NONE	No symptoms selected

4.3.4 Rights & Licence Section

A dedicated section explains to users in plain language: that no data is collected, the terms of the MIT licence, that health data is available in open formats, and the user's right to accurate health information and non-judgemental care.

4.3.5 SDG Alignment Display

A visual strip highlights the four UN Sustainable Development Goals this tool supports: SDG 3 (Good Health & Well-being), SDG 6 (Clean Water & Sanitation), SDG 10 (Reduced Inequalities), and SDG 17 (Partnerships for the Goals).

4.4 Accessibility and Usability

- A skip-to-content link is provided for keyboard and screen reader users.
- All navigation is keyboard-accessible with visible focus states styled in a high-contrast ochre colour.
- ARIA labels are applied to the ticker and the result panel (aria-live="polite"), so screen readers announce guidance changes automatically.

- The responsive grid layout adapts to mobile screens without requiring a separate mobile version.
- The tool was tested on low-cost Android devices to verify performance on typical Sierra Leonean consumer hardware.

5. Data Accessibility and Interoperability

A core requirement of Digital Public Goods is that data must be accessible, reusable, and interoperable with other systems.

In Salone Health Watch, all health topic content and symptom guidance rules are stored as plain JavaScript objects — effectively a JSON-like format — directly inside `script.js`. For example, the guidance object maps symptom combinations to plain-text advice messages, urgency levels, and headings.

This structure means:

- Any developer can extract the health content by copying the JavaScript object and saving it as a `.json` file, without any parsing tools.
- The data can be integrated into an SMS-based health bot (e.g. using RapidPro or Twilio), a mobile app, or a voice response system.
- Translations into Krio, Mende, Temne, or other Sierra Leonean languages can be done simply by replacing the English text values in the object, without touching any code logic.
- The topic card content in `index.html` uses semantic HTML, making it easily scraped or extracted by automated tools.

The tool uses no proprietary formats or vendor-locked data structures. All content is accessible through standard HTML, CSS, and JavaScript — open standards maintained by the World Wide Web Consortium (W3C).

6. Documentation and Repository

6.1 Project Files

File	Purpose
index.html	Main HTML structure of the application — all sections and content
style.css	All CSS styling, colour variables, layout, and responsive rules
script.js	Symptom guide JavaScript logic and guidance data
LICENSE	Full MIT License text
README.md	Project overview, features, installation, and contributing guide

6.2 README Overview

The README.md file documents: the project purpose and context, all features of the tool, installation and usage instructions (clone repository, open index.html), technology stack, data accessibility and interoperability approach, privacy and legal compliance, license information, ethical and accessibility considerations, contribution guidelines, and group attribution.

6.3 Version Control

The project is managed using Git and hosted on GitHub. Version control allows the team to track changes, resolve conflicts, and maintain a history of the project's development. The repository is publicly accessible under the MIT License.

7. SDG Alignment and Local Relevance

SDG 3 — Good Health and Well-being

Sierra Leone has one of the highest maternal mortality rates in the world and a high burden of malaria. Salone Health Watch directly addresses SDG 3 by providing community-accessible guidance on illness recognition, maternal care, vaccination, and outbreak awareness — all at no cost.

SDG 6 — Clean Water and Sanitation

Cholera outbreaks remain a recurring problem in Sierra Leone, particularly in urban informal settlements where access to safe water is unreliable. The tool's dedicated cholera and water safety topic card, linked to SDG 6, educates communities about water treatment and handwashing.

SDG 10 — Reduced Inequalities

By being free, open-source, offline-capable, and designed for low-cost devices, Salone Health Watch removes economic and technological barriers to health information. It serves both urban and rural populations equally, without requiring a smartphone with data connectivity.

SDG 17 — Partnerships for the Goals

The MIT License explicitly enables NGOs, clinics, the Ministry of Health, and international health organisations to reuse, adapt, and redistribute the tool. This creates the conditions for partnerships that extend the tool's reach beyond the immediate project.

8. Assessment Criteria Mapping

The following table maps each marking criterion to our project's specific achievements:

Criterion	Marks	Our Achievement
1. Relevance to SDGs & Sierra Leone Problem	/5	Addresses malaria, cholera, maternal health — all critical in SL. Aligned to SDGs 3, 6, 10, 17.
2. Open Source License Use	/5	MIT License applied, file included in repository, justification documented.
3. Technical Implementation	/15	Responsive HTML5/CSS3/JS single-file app; symptom logic with 7 urgency states; animated SVG visual; accessible markup.
4. Documentation & GitHub Quality	/5	README covers features, install steps, tech stack, privacy, contributing, and SDG alignment.
5. Data Accessibility & Interoperability	/10	Health data structured as plain JS objects (JSON-like); extractable for SMS bots or mobile apps.
6. Privacy & Legal Compliance	/5	Zero data collection; all processing client-side; MIT notice maintained; non-diagnostic disclaimer.
7. Standards & Best Practices	/10	Semantic HTML, ARIA labels, keyboard navigation, reduced-motion support, clean separated code.
8. Innovation & Creativity	/10	Animated ECG / Sierra Leone city map; scrolling health-alert ticker; three-level urgency colour coding.
9. Presentation & Report Quality	/10	Full written documentation (this document); structured presentation script prepared.
10. Teamwork & Collaboration	/5	Roles assigned; version-controlled on GitHub; README credits group.
11. Presentation	/10	Live demo planned; script prepared; Q&A coverage included.
12. Exhibition	/10	Tool runs on flash drive with no internet required; demonstrable on any device.
Total	100	

9. Conclusion

Salone Health Watch demonstrates that a meaningful, legally compliant, and technically sound Digital Public Good can be built with standard, open web technologies — no proprietary tools, no servers, and no cost to end users.

The project integrates the core principles of DLA207 — intellectual property law, open licensing, data privacy, and digital ethics — into a real application that addresses a genuine public health need in Sierra Leone. By applying the MIT License, structuring data for interoperability, and designing for accessibility, the tool meets both the academic requirements of this module and the international standards for Digital Public Goods.

We believe Salone Health Watch could be extended and adopted beyond this academic exercise — translated into Krio, integrated with SMS health bots, or adapted by the Ministry of Health — demonstrating the real-world value of applying IT law and ethical principles to software development.

10. References

- Digital Public Goods Alliance (2024). Digital Public Goods Standard. <https://digitalpublicgoods.net/standard/>
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